CS 255 System Design Document Template

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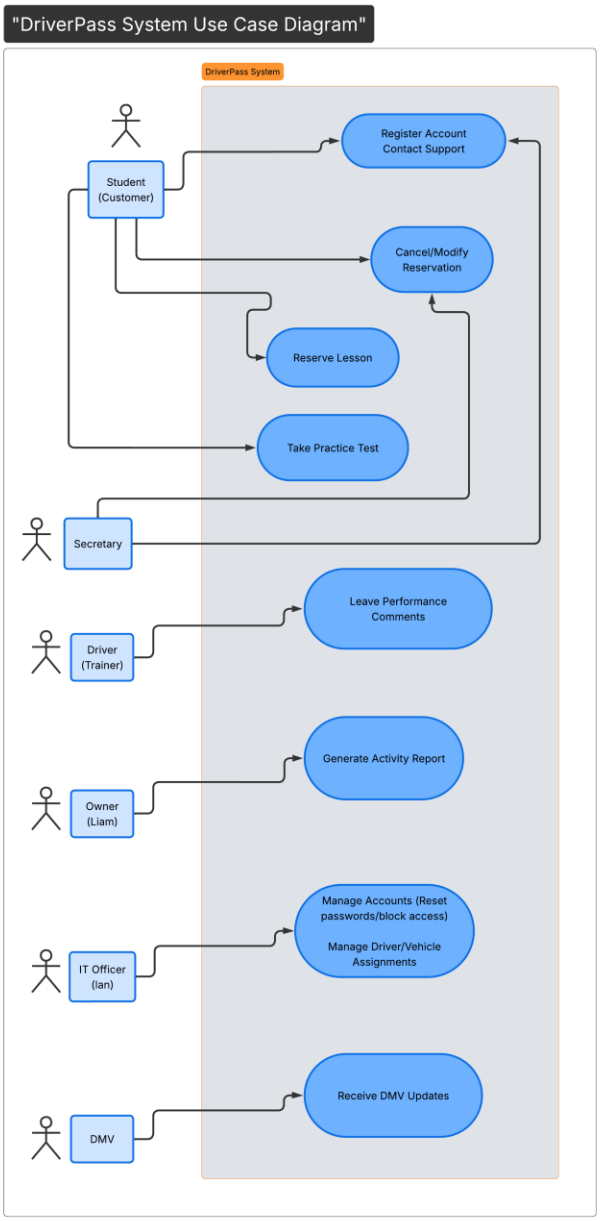
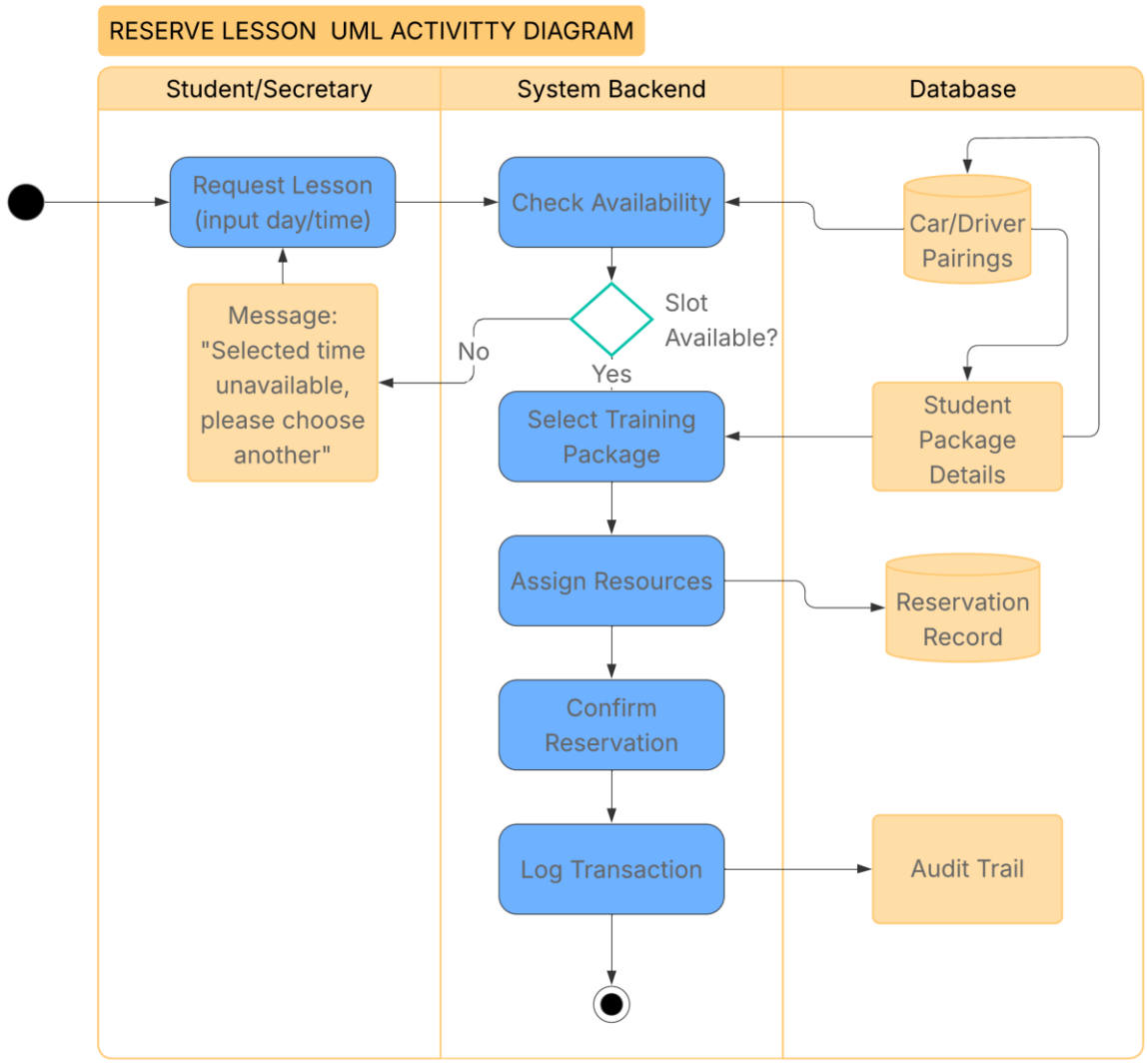
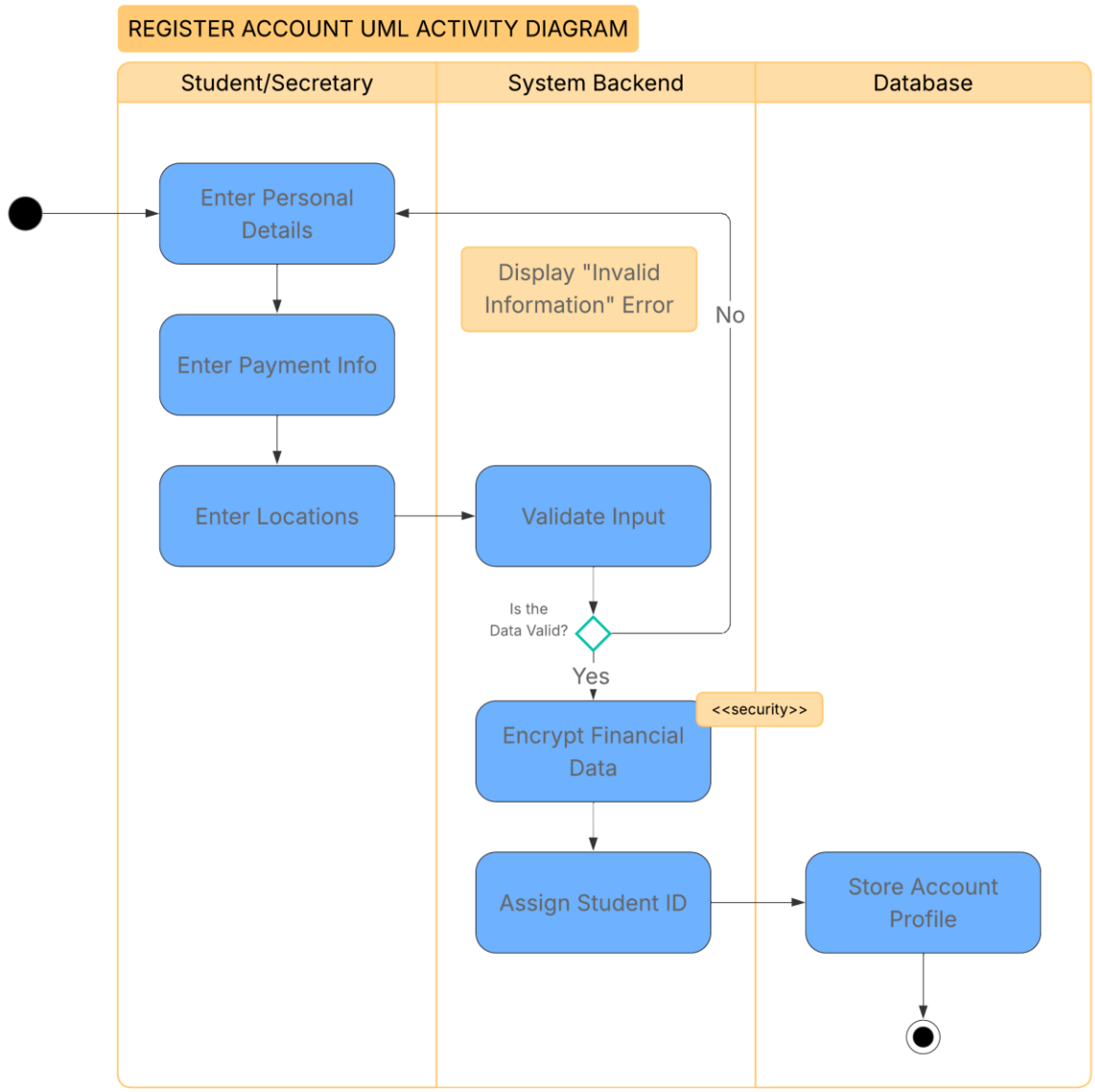
UML Diagrams

UML Use Case Diagram



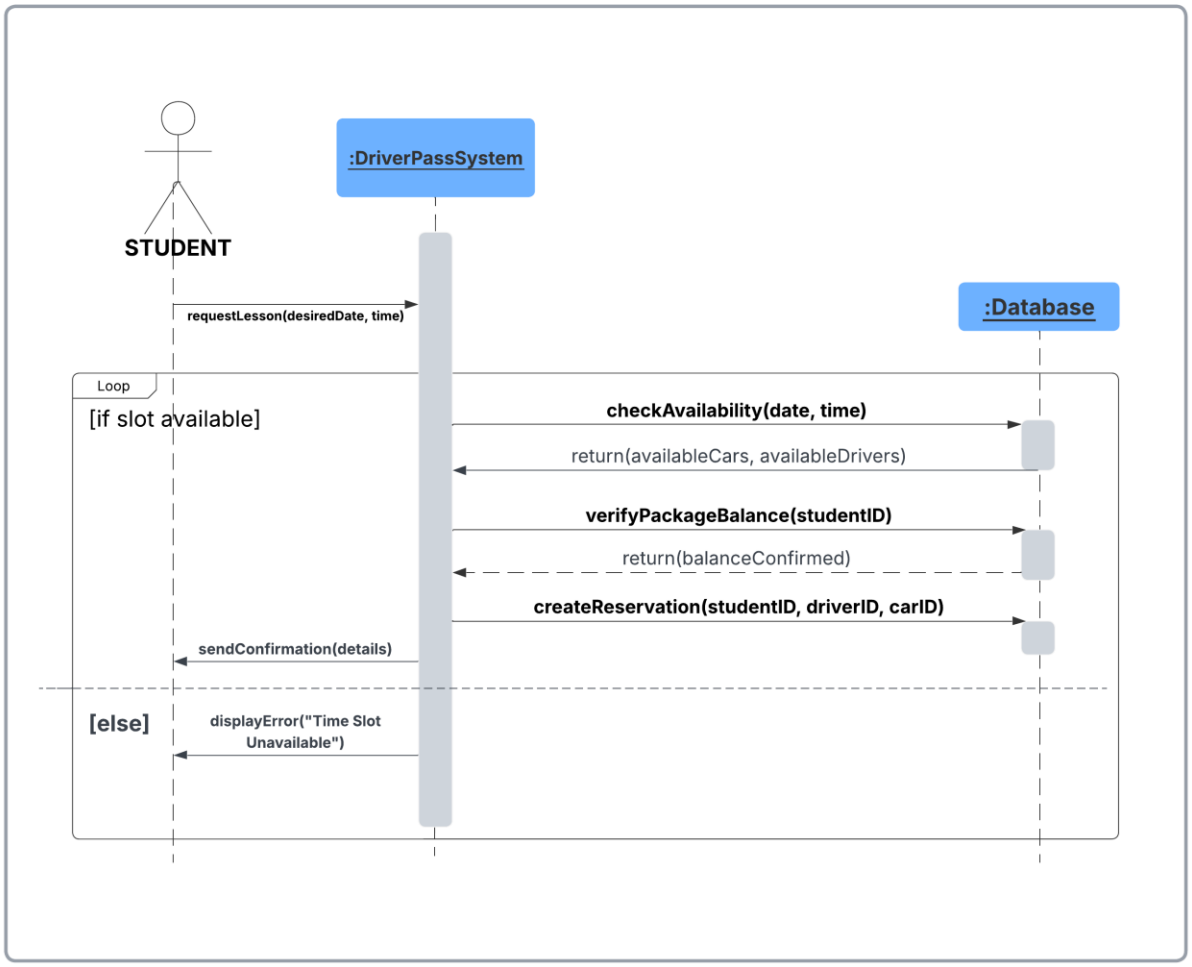
UML Activity Diagrams


RESERVE LESSON UML ACTIVITTY DIAGRAM
Student/Secretary
System Backend
Database
Request Lesson (input day/time)
Check Availability
Car/Driver Pairings
Message: "Selected time unavailable, please choose another"
No
Slot Available?
Yes
Select Training Package
Student Package Details
Assign Resources
Reservation Record
Confirm Reservation
Log Transaction
Audit Trail

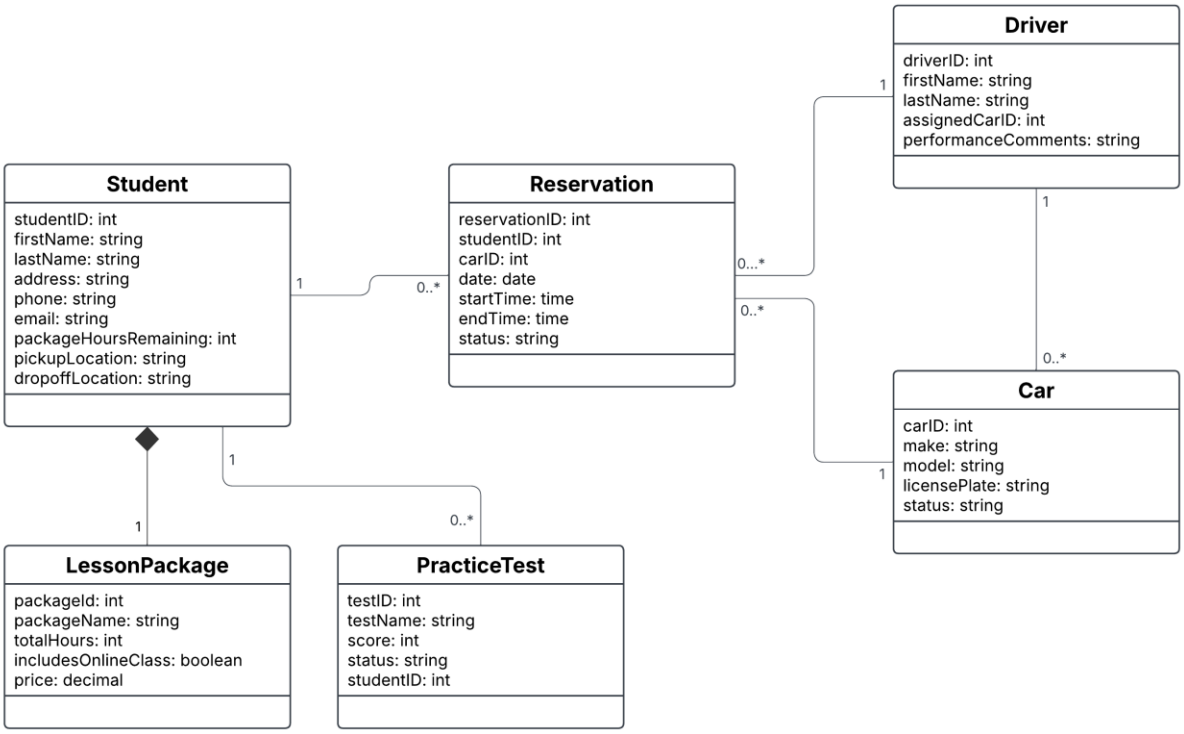


UML Sequence Diagram

Reserve Lesson Sequence diagram



UML Class Diagram



Technical Requirements

To bring the DriverPass system to life we need a setup that is both reliable and secure at the end of the day. Following Ian's lead on keeping things low maintenance the plan is to build this as a cloud native application. This will ensure Liam can focus on the business while the infrastructure handles the heavy lifting.

Infrastructure and Hosting

Cloud Platform: I'm proposing to host everything on AWS or Microsoft Azure. This gives a 24/7 web based environment without the headache of managing physical servers in an office and having to worry about downtime...

Application Hosting: Using managed services like AWS Fargate or Azure App Service will let the system scale easily. If Liam decides to double his fleet from 10 to 20 cars the backend can handle the extra traffic without missing a beat and having bottlenecks.

Database: We'll use a managed relational database something like Amazon RDS (running PostgreSQL) is ideal. This is perfect for managing the complex links between students, drivers, and those two hour lesson slots I mapped out in the Class Diagram...

Storage: For things like student/driver profile photos and those monthly activity reports Liam wants we can use Amazon S3 for secure and long term cloud storage.

Software and Application Stack

Development Framework: I would definitely recommend a modern stack like Node.js or Python (Django). These are great for handling the "heavy" logic required to check availability across multiple cars and drivers in real time.

Reporting & Exports: To meet the requirement for offline work the system will have to use a library like ExcelJS. This lets Liam download his data directly into Microsoft Excel so he can run his own numbers at home.

Notifications: Also we'll integrate an SMTP service to trigger automatic emails. So when the moment a student finishes booking a lesson, the system will fire off a confirmation to both the student and their assigned driver to keep track of progress on both ends...

Hardware Requirements

Office Setup: The Secretary just needs a standard desktop or laptop (at least 8GB of RAM) and a stable internet connection this is a given. This will just be the primary hub for handling those phone-in appointments and managing the daily schedule.

Field Equipment: Since the training happens on the road, drivers will use 4G/5G-enabled tablets or smartphones. This lets them check the student's photo for safety, log their lesson hours, and leave performance notes the second the session ends we could possibly include mobile hotspots to enable connection while on the road.

Fleet Tech (Future Phase): Down the road we could equip the 10 cars with GPS. While not required now it would allow the system to automatically sync pickup and drop-off locations with the actual lesson logs.

Security and External Integration

Access Control: We'll implement Role Based Access Control like IAM. This is a big one for Ian it ensures students only see their own progress while he has the "admin" power to reset passwords or block access if an employee leaves.

Data Security: All sensitive info, especially credit card numbers, will be protected with AES-256 encryption. We'll also use SSL/TLS certificates so that every bit of data moving between the user and the cloud is encrypted and safe this is going to be essential for sensitive data handling like this..

DMV Integration: Finally the system needs a secure API link to the DMV. This way, DriverPass gets a "ping" or notification the second the DMV updates their rules or adds new practice test questions.